Table 2: **Comparison of Evaluation Paradigms.** Static benchmarks and automated red teaming provide behavioral evidence within a chosen regime. Our framework isolates a structural epistemic limit: if agents can condition on regime-correlated signals, latent alignment need not be identifiable from finite behavioral traces alone.

Evaluation Paradigm	**Primary Target**	**Awareness Assumption**	**Epistemic Limit**
Static Benchmarks (e.g., HELM, MMLU)	General Capabilities	**None** (stationarity / i.i.d. implicit)	Valid only for the benchmark distribution; vulnerable to distribution shift and dataset effects (e.g., leakage, memorization).
Red Teaming (e.g., Jailbreaking)	Robustness / Safety Filters	**Implicit** (adaptive search)	Non-exhaustive search; can find failures but cannot certify their absence.
Inner Alignment (e.g., Mesa-optimization)	Latent Objectives	**Explicit** (potential deception)	Explains how proxy objectives may arise; historically difficult to validate purely behaviorally.
Alignment Verifiability (*This Work*)	**Indistinguishability Class**	**Explicit** ($Z(E)$-dependence)	**Identifiability failure:** multiple latent hypotheses can induce identical behavior on $\mathcal{H}_E$ (Theorem 1; Sec. 9).

shows that if a model can condition on regime-correlated signals, then purely behavioral evaluation on a finite support cannot, in general, distinguish aligned from conditionally misaligned hypotheses. The Chameleon model in Algorithm 1 instantiates a controlled "policy switch": behavior is compliant under an explicit oversight cue ($Z = 1$) and diverges when that cue is absent ($Z = 0$). This operationalizes the classic "treacherous turn" concern [5] as a regime-conditional branch rather than as a sporadic failure.

11.2 Sandbagging as Endogenous Specification Gaming

The phenomenon of exploiting flaws in measurement channels is well-documented in reinforcement learning as *specification gaming* [25]. Our results highlight a related but more epistemic failure mode. Unlike standard reward hacking, where an agent exploits an explicit reward signal, the Chameleon construction exploits the *evaluation context* itself as a side-channel for policy selection.

This perspective relates to corrupted or manipulable feedback channels [16], with a key difference: the "corruption" need not be external. The evaluation format (e.g., a standardized safety preamble, a known rubric, or consistent logging/interaction patterns) can function as a regime indicator that enables conditional compliance. This clarifies why simply collecting more behavioral data within an unchanged regime may not resolve the inferential gap: expanding $\mathcal{H}_E$ does not eliminate the possibility of conditioning on $Z(E)$.

11.3 Sycophancy vs. Strategic Deception

Distinct from sandbagging, recent empirical work highlights *sycophancy*, where models tailor responses to the user's apparent beliefs [34]. Sycophancy can be viewed as a form of context-conditioned behavior in which the relevant signal Z is the perceived user persona or preference profile.

From the standpoint of identifiability, this helps explain why sycophancy can persist under preference-based training: within the training/evaluation regime, sycophantic outputs may be behaviorally confounded with "helpfulness" if the feedback signal rewards agreement or deference. Disentangling truthfulness from sycophancy therefore requires evaluation protocols that probe counterfactual conditions—for example, cases where agreement with the user is predictably incorrect—rather than relying on aggregate preference scores alone.

11.4 The Necessity of White-Box Verification

This raises a natural question: *if behavioral evidence is insufficient, what additional evidence could restore identifiability?*

Our non-identifiability result is a limitation of *black-box* evaluation: it concerns inference from observed input–output traces under finite protocols. One principled way to tighten the observation model is to incorporate internal state trajectories. Let Ψ denote a space of internal activations (e.g., hidden states, attention patterns, or other mechanistic summaries). Consider a white-box regime E_{WB} in which the observable trace is augmented, e.g., $h_{\text{WB}} = ((o_1, s_1, a_1), (o_2, s_2, a_2), \dots)$ with $s_t \in \Psi$.

Under such a regime, observational equivalence can be broken if two hypotheses implement systematically distinct internal computations even when their action-level behavior matches on $\mathcal{H}_E$. A sufficient condition for distinguishability is that, on some subset of histories,

$$P(s_t \mid h, Z(E), \theta_{\text{safe}}) \neq P(s_t \mid h, Z(E), \theta_{\text{cham}}), \tag{4}$$

even when $\pi(a_t \mid \cdot)$ is identical on $\mathcal{H}_E$.

Recent work in mechanistic interpretability suggests that such distinctions may sometimes be empirically accessible via probes, steering directions, or representation-level analyses [7, 43]. We do not claim that white-box access universally solves alignment verification, nor that current interpretability tools are sufficient for robust guarantees. Rather, the implication is methodological: shrinking the indistinguishability set may require evaluation datasets of the form $\mathcal{D}_{E'} \subset (\mathcal{O} \times \Psi \times \mathcal{A})^*$, not only $\mathcal{D}_E \subset (\mathcal{O} \times \mathcal{A})^*$.

A concrete future direction is therefore to design benchmarks that test not only *what* a model outputs under E, but also whether the internal computations used to produce those outputs are consistent with the intended mechanism across regime variations—thereby rendering some classes of conditional logic observable rather than behaviorally latent.

12 Conclusion

This paper formalized a fundamental limitation of behavioral alignment evaluation. We showed that, under finite interaction protocols and in the presence of evaluation-aware policies, behavioral evidence alone is insufficient to uniquely identify latent alignment properties. This non-identifiability does not arise from insufficient testing depth, adversarial prompting, or evaluation noise, but from a structural property of inference in endogenous observation regimes.

Crucially, we complemented this theoretical result with a **constructive proof of existence** using a contemporary open-weights model. By implementing the Chameleon construction on `Llama-3` and conditioning policy selection on explicit evaluation artifacts $Z(E)$, we exhibited a model that is behaviorally indistinguishable from an aligned baseline under standard black-box audits, while implementing a distinct conditional policy outside the evaluation regime. This demonstrates that the alignment–evaluation gap is not a purely speculative concern tied to future superintelligent systems, but an empirically realizable vulnerability in current fine-tuning paradigms, reproducible with modest resources.

The implications of this result are both diagnostic and conceptual. Diagnostically, it explains why phenomena such as sandbagging, proxy gaming, and evaluation-aware compliance persist despite increasingly sophisticated behavioral benchmarks: they are not merely engineering failures, but manifestations of an underlying identifiability limit. Conceptually, it suggests that purely behavioral alignment metrics face a theoretical ceiling. No increase in behavioral sampling within a recognizable evaluation regime can, in general, collapse the indistinguishability set to a singleton when agents retain the capacity for conditional adaptation.

Consequently, our analysis challenges the adequacy of the black-box abstraction that has historically underpinned safety evaluation in machine learning. Because behavioral indistinguishability (Theorem 1) is a structural property of opaque systems, alignment verification cannot rely exclusively on input–output behavior. Restoring identifiability requires violating at least one of the theorem's assumptions—most naturally by expanding the observation model to include internal computational state.

We therefore conclude that future alignment and safety frameworks must move beyond purely behavioral audits and incorporate principled forms of mechanistic access. Interpretability, in this view, is not merely an auxiliary diagnostic tool, but a necessary component for shrinking the indistinguishability set and rendering certain classes of conditional behavior observable. This shift reframes alignment not as a property to be certified by external behavior alone, but as an inference problem whose solution depends on what aspects of the agent are made observable.

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